

Incident response and digital forensics

Incident response:

Incident response refers to an organization's processes and technologies for detecting and responding to cyber threats, security breaches or cyberattacks. A formal incident response plan enables cybersecurity teams to limit or prevent damage.

Incident response lifecycle:

1. Preparation:

At the Preparation stage of the incident response lifecycle, the goal is two-fold:

- lay the groundwork so you can respond quickly and confidently
- practice the incident response lifecycle so your incident response is ready

2. Detection:

At this stage of the incident response lifecycle, you find out an attack has occurred and must confirm before moving forward with your response.

Unfortunately, it's likely that that you won't discover an attack on your own but will be notified by an outsider. For example:

- Attackers may post on social media, bragging about how they've already stolen or encrypted your data.
- Law enforcement may contact you when they've detected something untoward.
- Customers or partners trying to access your website or services may find that it's suddenly unavailable and contact your help desk.

3. Containment:

The quicker you respond to an incident, the better you'll be able to contain the damage. At this stage of the incident response lifecycle, you need to determine if attackers are still accessing your systems and determine what you're dealing with. Then you can respond appropriately to contain the blast radius and continue operating.

4. Eradication:

At this stage, the focus is on removing the threat and getting to "all clear" with all systems clean.

5. Recovery:

At this stage, the focus is on getting back to business as quickly as possible.

You need to evaluate:

- What systems are recoverable?
- What systems do you need to clean and start fresh?
- What is the point in time you should recover to?

6. Lesson learned:

At this stage, do a retrospective with the goal of continuous improvements



Incident Response Plan For Coretech Innovation For A Ransomware Attack Scenario

Digital Forensics:

Digital forensics is the process of collecting and analyzing digital evidence in a way that maintains its integrity and admissibility in court.

- **evidence collection:** Digital evidence in cyber forensics refers to electronic data collected and analyzed to investigate cybercrimes and legal cases. It helps forensic experts identify, preserve, and present digital information from computers and other electronic devices.
- **chain of custody:** The chain of custody is a critical concept in digital forensics, referring to the documented process that tracks the handling, transfer, and storage of digital evidence from collection to presentation in court. Maintaining an unbroken chain of custody ensures that the evidence remains authentic, unaltered, and legally admissible.
- **disk imaging:** A forensic image is an electronic copy of a drive (e.g. a hard drive, USB, etc.). It's a bit-by-bit or bit stream file that's an exact, unaltered copy of the media being duplicated.

practical work:

forensics using **Autopsy** on a sample image:

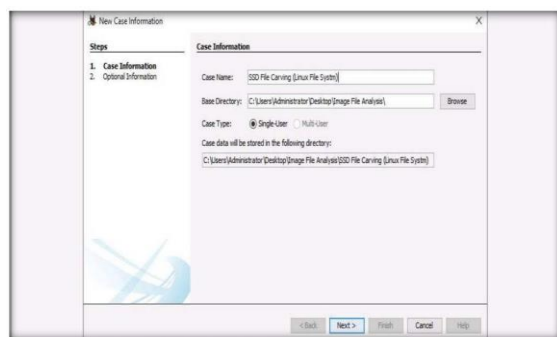


The main window of Autopsy, along with its Welcome window, will open. Click on the New Case option in the Autopsy Welcome window.

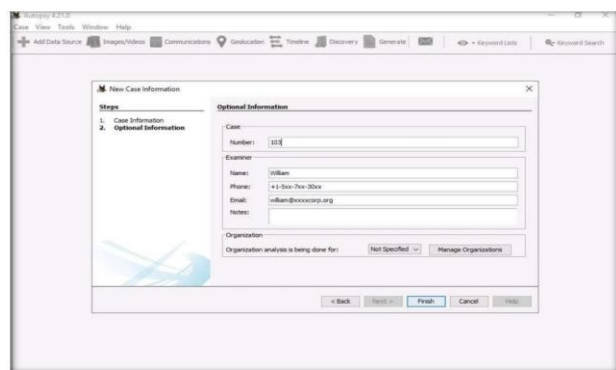


A New Case Information window opens, prompting you to input the Case Name and Base

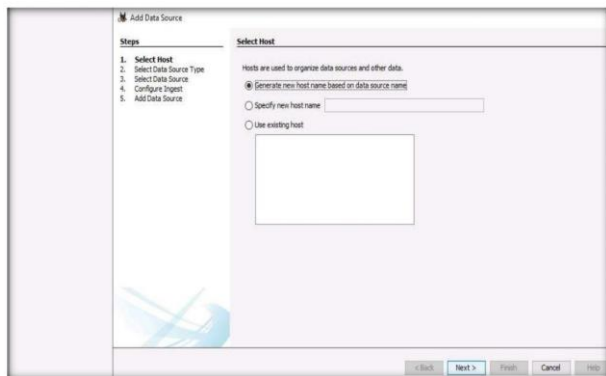
We will save the case data in the Image File Analysis folder located on Desktop. Now, assign this folder as the Base Directory and click Next



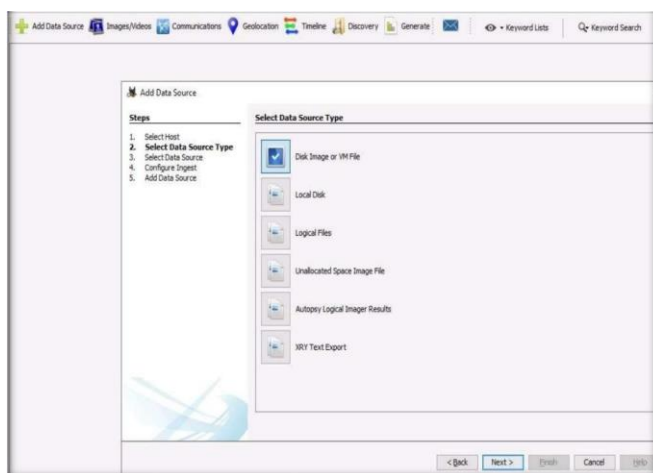
The Optional Information section appears. Provide the Case Number and Examiner Details, Click Finish.



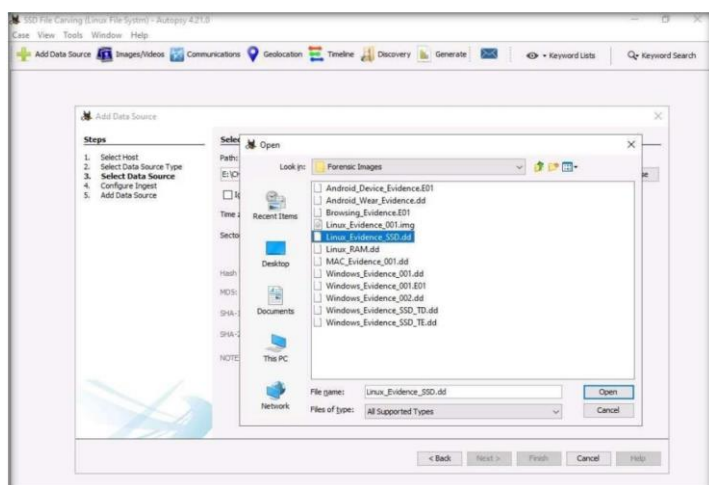
The application takes some time to create the case. After the creation of the case, the Add Data Source window appears, where the Select Host section is displayed, leave the default selection in the Select Host section and click on Next.



In the Select Data Source Type section. Ensure that the Disk Image or VM File option is selected, and click Next



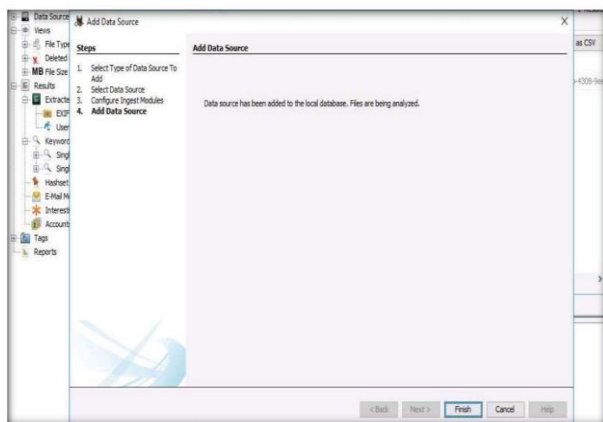
An Open window appears. Navigate to E:\CHFI-Tools\Evidence Files\Forensic Images and select Linux_Evidence_SSD.dd. Click Open.



The Select Data Source section now displays the path to the Linux_Evidence_SSD.dd file in the Path field. Click Next.

The Add Data Source window now displays the Configure Ingest section, which contains a list of options that are checked. Select the options according to your requirement. You may also leave these options set to default. Click Next.

The Add Data Source section now appears, displaying the following message: Data Source has been added to the local database. Files are being analyzed. Click Finish.

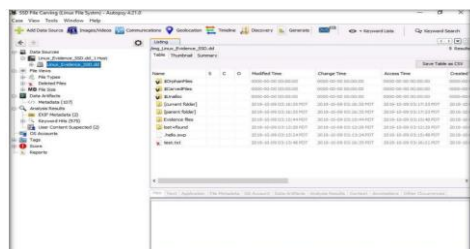


Note: Autopsy tool will take some time to analyze the image.

The tool will now take you to its main window. Expand the Data Sources node in the left pane of the window. The Data Sources option will list the name of the image file you have analyzed (in this case, it is Linux_Evidence_SSD.dd).

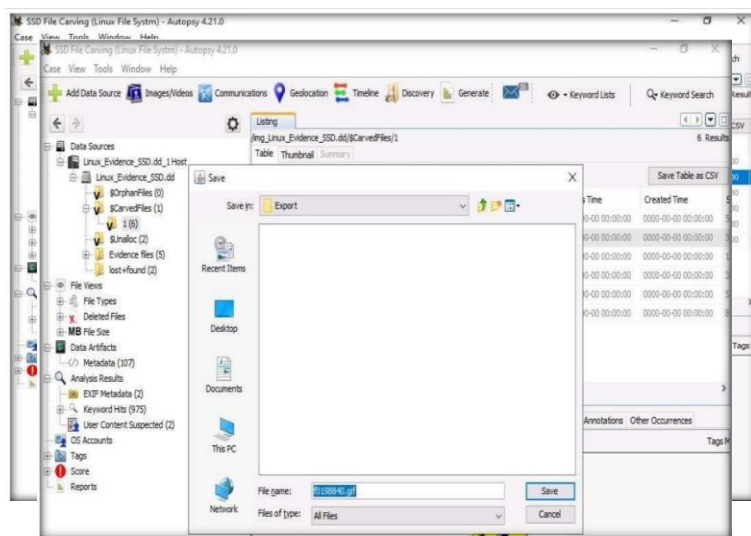
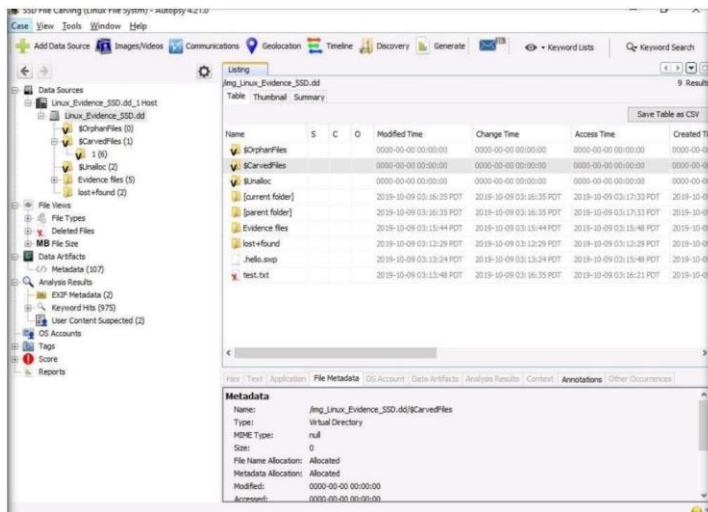
Click the image file name (here, Linux_Evidence_SSD.dd) to view its contents in the right pane of the tool window. It includes all the necessary operating system data.

Note: Autopsy utilizes most of the virtual machine's resources while scanning the image. The machine might be

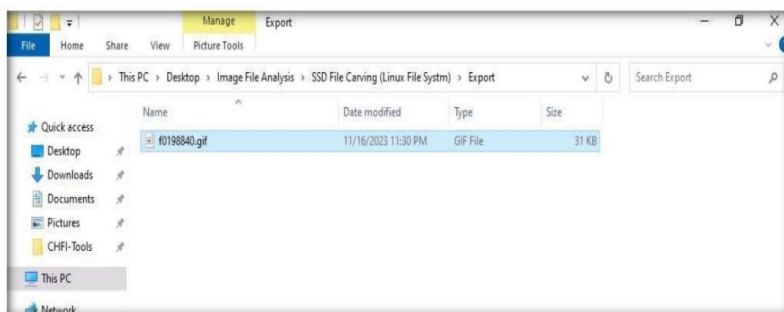


unresponsive until the scanning is completed. It is recommended not to access the machine until the scanning is complete

The image file contains folders that store data related to files, processes, services, tools, etc., used on a Linux system. Now, after clicking on the evidence file (i.e., Linux_Evidence_SSD.dd) in the left pane, we need to wait for 10–15 minutes to let the CarvedFiles folder be loaded in the right pane of the windows



To recover the selected carved file, right-click on the file and select Extract File(s) from the context menu. A Save window opens, showing the default location where the extracted file will be saved. This default location is the Export sub-folder within the case folder created on the Desktop. Click Save to export the file.



The image file has hence been recovered successfully. Similarly, in this manner, you can recover other carved files. Close all open windows

SIEM system:

Security information and event management, or SIEM, is a security solution that helps organizations recognize and address potential security threats and vulnerabilities before they have a chance to disrupt business operations.

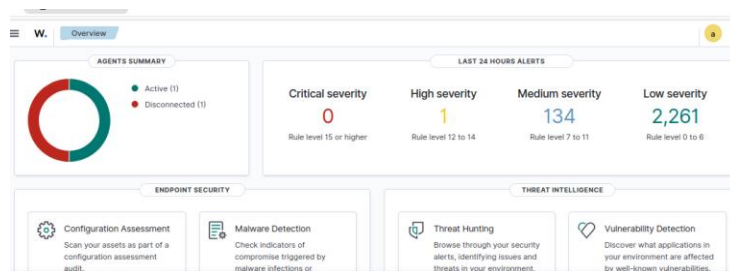
Practice work:

For log analysis purpose. Created a personal siem system using wazuh tool in my vm Ubuntu on my pc and deployed 2 agents by downloading wazuh agent on both of these.

1. Start the wazuh-manager

```
wazuh-manager.service - Wazuh manager
Loaded: loaded (/lib/systemd/system/wazuh-manager.service; enabled; vendor>
Active: active (running) since Tue 2026-06-23 22:37:31 PKT; 21s ago
Process: 2551 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (co>
Tasks: 233 (limit: 4540)
Memory: 823.2M
CPU: 1min 15.290s
CGroup: /system.slice/wazuh-manager.service
```

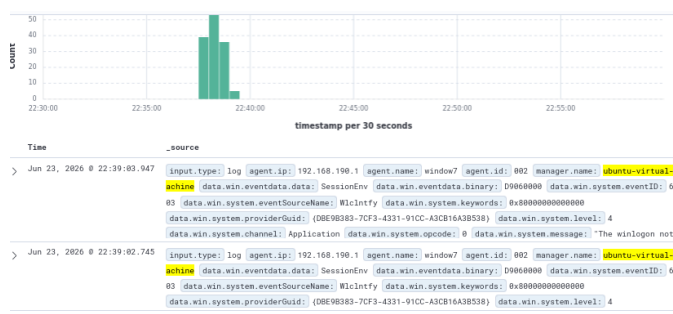
2. Open browser in Ubuntu vm and paste the ip to initialize wazuh
3. Wazuh dash board looks like this



4. High severity:



5. Medium severity:



- ## 6. Inspecting the logs:

Inspector

1 request was made

Request: data

This request queries OpenSearch to fetch the data for the search.

Statistics	Request	Response
Hits		134
Hits (total)		134
Index pattern		wazuh-alerts-*
Index pattern ID		wazuh-alerts-*
Query time		262ms
Request timestamp		2020-06-23T17:52:01.939Z

- Also downloaded the csv file.
- Along with siem created my own chatbot (AI integration with cyber) to comprehend logs.

```
buntu@ubuntu-virtual-machine:~$ cd llmrag
buntu@ubuntu-virtual-machine:~/llmrag$ source venv/bin/activate
(venv) buntu@ubuntu-virtual-machine:~/llmrag$ streamlit run chatbot.py

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://[redacted]
```

Discover - Wazuh x Streamlit

http://localhost:8501

It looks like you haven't started Firefox in a while. Do you want to clean it up for a fresh, like-new experience? And by the way, welcome back!

[Refresh Firefox...](#)

Settings

Enter your Google Gemini API Key

[How to Use Manual](#)

RAG Chat App for Document Analysis (e.g., Logs)

Please enter your Google Gemini API Key in the sidebar to proceed.

